

Week 1: Interim Deliverable

Short Paper Notes:

1. *Tyler et al. (2024)*

Tyler et al. reviewed recent research on the use of artificial intelligence in emergency department triage. The clinical problem is that emergency departments face overcrowding, variable triage decisions, and pressure to identify high-risk patients quickly. The review found that AI and machine-learning tools may support acuity prediction, resource allocation, and earlier recognition of patients needing urgent care. However, the authors highlight that much of the evidence is still early-stage, with many studies relying on retrospective data rather than live clinical testing.

2. *Da'Costa et al. (2025)*

Da'Costa et al. reviewed the benefits, challenges, and future direction of AI-driven triage in emergency departments. The paper explains that AI could improve triage speed, consistency, and decision support, particularly in busy departments where clinicians must make rapid judgements with limited information. It also identifies concerns around bias, data quality, explainability, privacy, and clinician trust. A key limitation is that strong technical performance does not automatically prove that an AI tool will be safe, fair, or useful in real emergency department workflows.

3. *Chen et al. (2024)*

Chen et al. developed and tested machine-learning and natural language processing models to predict patient disposition using structured triage data and free-text clinical notes. The problem addressed was that early triage decisions are made before the full clinical picture is available, while useful information may already be present in short nurse-written notes. Their models performed better than emergency physicians and a baseline logistic regression model for some disposition predictions. The authors note that their work focused mainly on immediate disposition outcomes and did not fully address longer-term outcomes, resource use, waiting time, or survival.

4. *Ottenhoff et al. (2023)*

Ottenhoff et al. studied whether machine learning could be used as an adjunct to traditional emergency department triage. The clinical problem was that standard triage systems can miss patients whose risk becomes clearer only after combining multiple pieces of early clinical information. Their work suggests that machine-learning models may help identify patients at risk of adverse outcomes and support more consistent triage decisions. A key limitation is that model performance alone does not show whether clinicians will trust the tool, use it correctly, or benefit from it in a live triage environment.

5. *Townsend et al. (2023)*

Townsend et al. explored medical practitioner perspectives on the use of artificial intelligence in emergency triage. The problem addressed was not only whether AI can predict risk, but whether clinicians believe it can fit safely into urgent clinical decision-making. The study found that practitioners saw potential value in AI support, especially for consistency and risk recognition, but raised concerns about accountability, transparency, bias, and over-reliance. This paper is useful because it shows that implementation challenges are not just technical; they also depend on clinician trust, workflow design, and clear responsibility for final decisions .

Two Literature Gaps:

1. Many AI triage studies report promising retrospective model performance, but fewer test whether these tools improve live nurse triage decisions in real emergency department workflows.
2. Several studies focus on accuracy, but fewer evaluate fairness, alert burden, explainability, and whether clinicians can safely understand and act on AI recommendations.

Draft Problem Statement:

Mercer General Hospital's ED faces risk of patient under-triage during peak periods, where nurse workload and time pressure can cause high-risk cases to be missed. A 12-week pilot will test an explainable AI support tool that uses structured triage data and free-text presenting complaints to flag high-risk cases for nurse review, displaying a visible reason for each alert. The pilot will measure sensitivity, specificity, alert burden, and fairness across demographic groups, with nurses retaining full accountability for all final escalation decisions.

Word Count: 88 words

References:

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Townsend, B. A., et al. (2023). Medical practitioner perspectives on AI in emergency triage. *Frontiers in Digital Health*, 5, 1297073. <https://doi.org/10.3389/fdgth.2023.1297073>